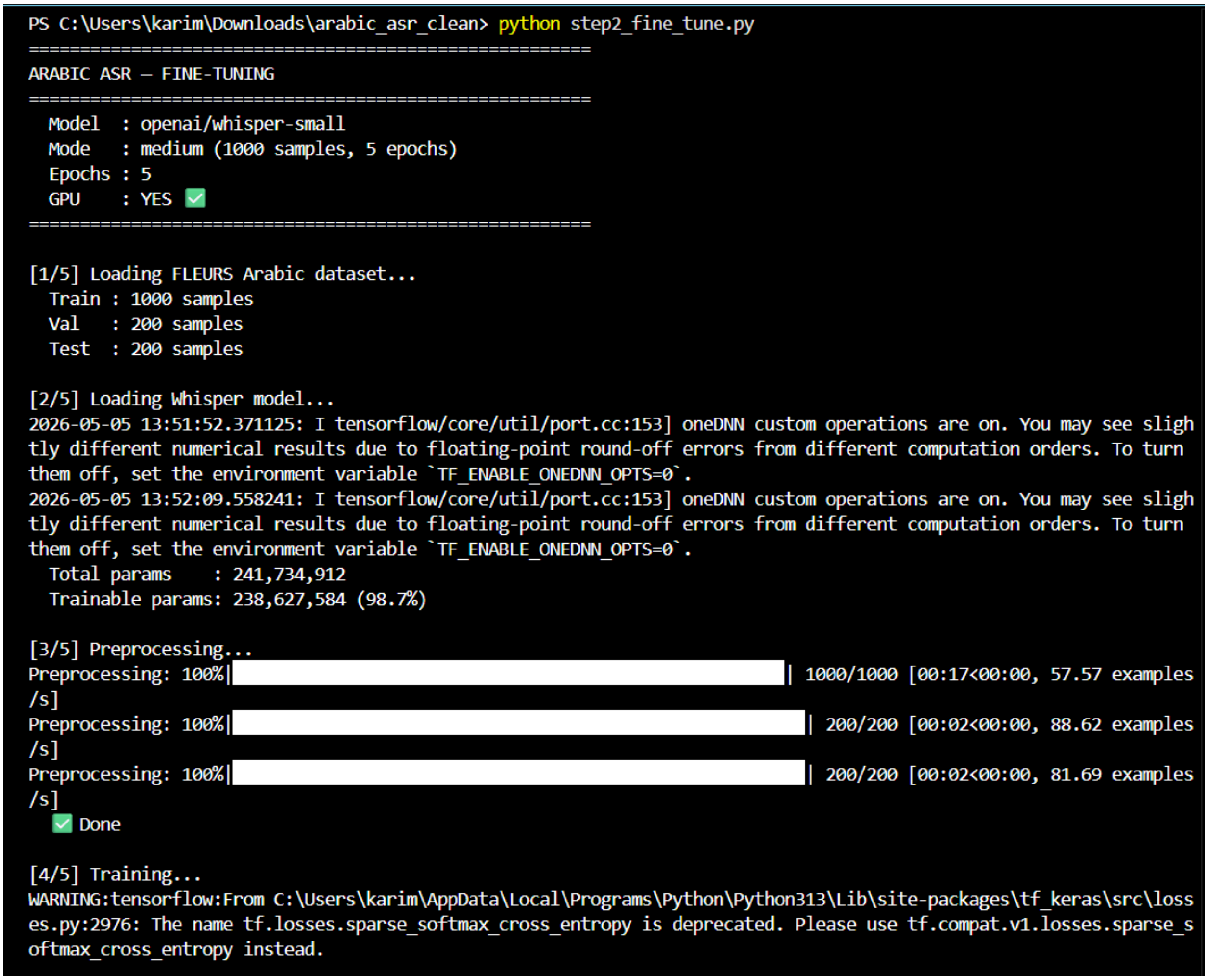
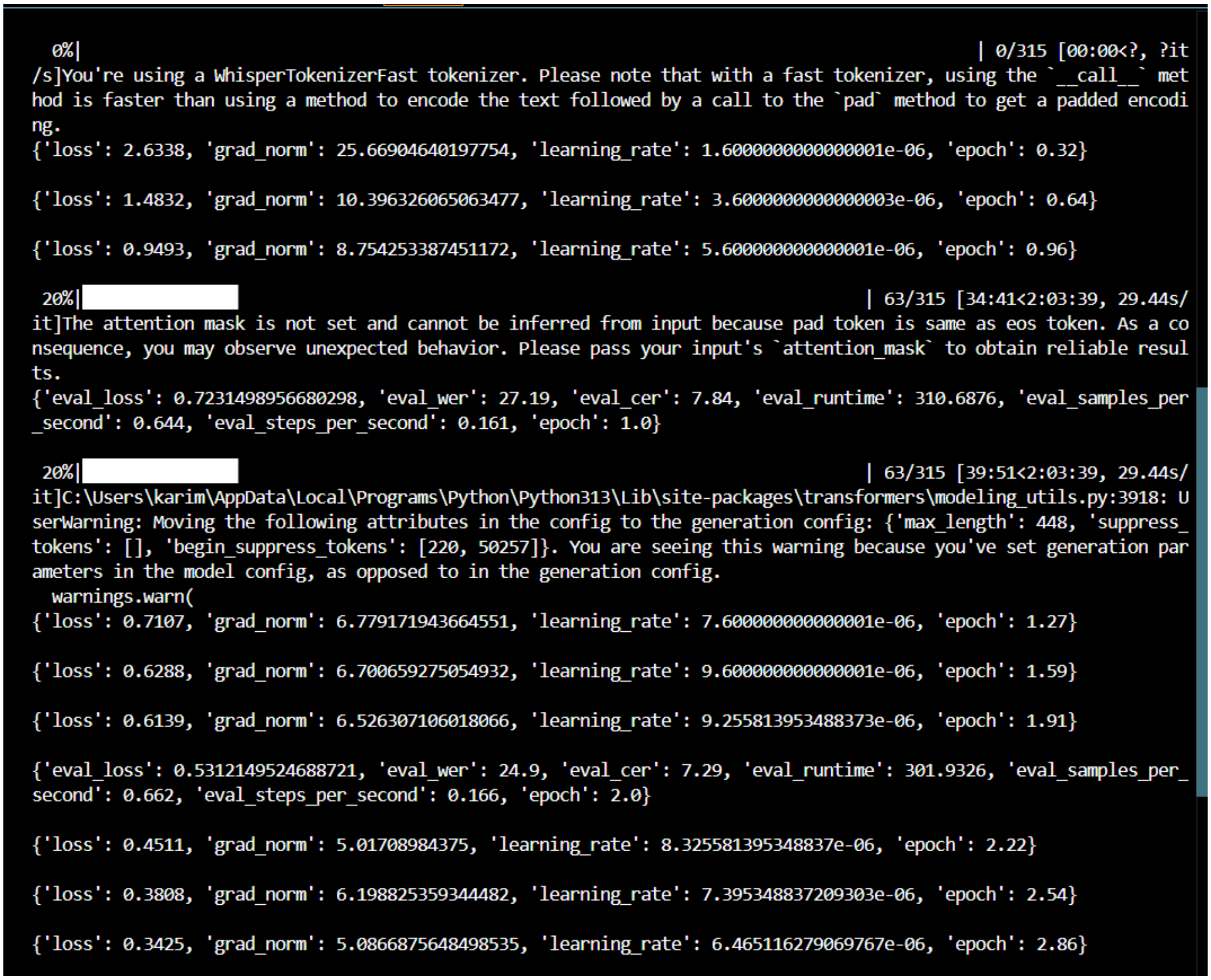
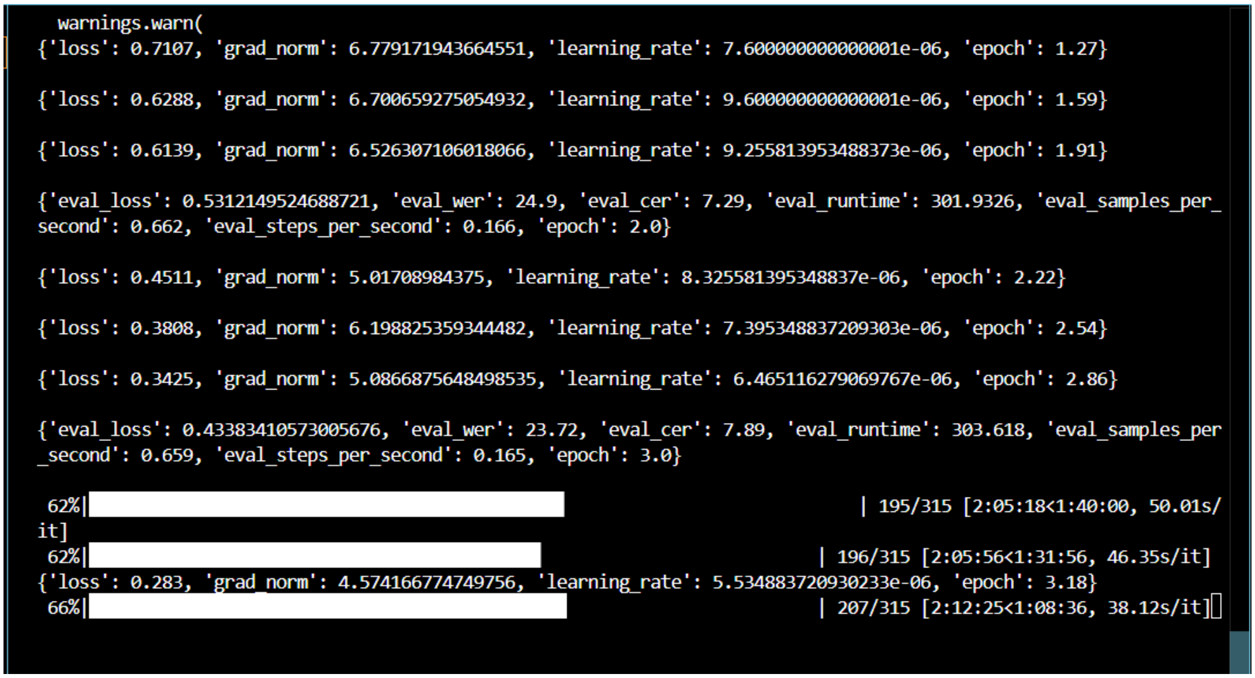


```
PS C:\Users\karim\Downloads\arabic_asr_clean> python step2_fine_tune.py
========================================================
ARABIC ASR — FINE-TUNING
========================================================
  Model   : openai/whisper-small
  Mode    : medium (1000 samples, 5 epochs)
  Epochs : 5
  GPU     : YES ✅
========================================================

[1/5] Loading FLEURS Arabic dataset...
  Train : 1000 samples
  Val    : 200 samples
  Test   : 200 samples

[2/5] Loading Whisper model...
2026-05-05 13:51:52.371125: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see sligh
tly different numerical results due to floating-point round-off errors from different computation orders. To turn
them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-05 13:52:09.558241: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see sligh
tly different numerical results due to floating-point round-off errors from different computation orders. To turn
them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
  Total params     : 241,734,912
  Trainable params: 238,627,584 (98.7%)

[3/5] Preprocessing...
Preprocessing: 100%|██████████████████████████████████████████████████████| 1000/1000 [00:17<00:00, 57.57 examples
/s]
Preprocessing: 100%|████████████████████████████████████████████████████████| 200/200 [00:02<00:00, 88.62 examples
/s]
Preprocessing: 100%|████████████████████████████████████████████████████████| 200/200 [00:02<00:00, 81.69 examples
/s]
  ✅ Done

[4/5] Training...
WARNING:tensorflow:From C:\Users\karim\AppData\Local\Programs\Python\Python313\Lib\site-packages\tf_keras\src\loss
es.py:2976: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses.sparse_s
oftmax_cross_entropy instead.
```



```
  0%|                                                                   | 0/315 [00:00<?, ?it
/s]You're using a WhisperTokenizerFast tokenizer. Please note that with a fast tokenizer, using the `__call__` met
hod is faster than using a method to encode the text followed by a call to the `pad` method to get a padded encodi
ng.
{'loss': 2.6338, 'grad_norm': 25.66904640197754, 'learning_rate': 1.6000000000000001e-06, 'epoch': 0.32}

{'loss': 1.4832, 'grad_norm': 10.396326065063477, 'learning_rate': 3.6000000000000003e-06, 'epoch': 0.64}

{'loss': 0.9493, 'grad_norm': 8.754253387451172, 'learning_rate': 5.600000000000001e-06, 'epoch': 0.96}

 20%|███████████████                                            | 63/315 [34:41<2:03:39, 29.44s/
it]The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a co
nsequence, you may observe unexpected behavior. Please pass your input's `attention_mask` to obtain reliable resul
ts.
{'eval_loss': 0.7231498956680298, 'eval_wer': 27.19, 'eval_cer': 7.84, 'eval_runtime': 310.6876, 'eval_samples_per
_second': 0.644, 'eval_steps_per_second': 0.161, 'epoch': 1.0}

 20%|███████████████                                            | 63/315 [39:51<2:03:39, 29.44s/
it]C:\Users\karim\AppData\Local\Programs\Python\Python313\Lib\site-packages\transformers\modeling_utils.py:3918: U
serWarning: Moving the following attributes in the config to the generation config: {'max_length': 448, 'suppress_
tokens': [], 'begin_suppress_tokens': [220, 50257]}. You are seeing this warning because you've set generation par
ameters in the model config, as opposed to in the generation config.
  warnings.warn(
{'loss': 0.7107, 'grad_norm': 6.779171943664551, 'learning_rate': 7.600000000000001e-06, 'epoch': 1.27}

{'loss': 0.6288, 'grad_norm': 6.700659275054932, 'learning_rate': 9.600000000000001e-06, 'epoch': 1.59}

{'loss': 0.6139, 'grad_norm': 6.526307106018066, 'learning_rate': 9.255813953488373e-06, 'epoch': 1.91}

{'eval_loss': 0.5312149524688721, 'eval_wer': 24.9, 'eval_cer': 7.29, 'eval_runtime': 301.9326, 'eval_samples_per_
second': 0.662, 'eval_steps_per_second': 0.166, 'epoch': 2.0}

{'loss': 0.4511, 'grad_norm': 5.01708984375, 'learning_rate': 8.325581395348837e-06, 'epoch': 2.22}

{'loss': 0.3808, 'grad_norm': 6.198825359344482, 'learning_rate': 7.395348837209303e-06, 'epoch': 2.54}

{'loss': 0.3425, 'grad_norm': 5.0866875648498535, 'learning_rate': 6.465116279069767e-06, 'epoch': 2.86}
```



```
  warnings.warn(
{'loss': 0.7107, 'grad_norm': 6.779171943664551, 'learning_rate': 7.600000000000001e-06, 'epoch': 1.27}

{'loss': 0.6288, 'grad_norm': 6.700659275054932, 'learning_rate': 9.600000000000001e-06, 'epoch': 1.59}

{'loss': 0.6139, 'grad_norm': 6.526307106018066, 'learning_rate': 9.255813953488373e-06, 'epoch': 1.91}

{'eval_loss': 0.5312149524688721, 'eval_wer': 24.9, 'eval_cer': 7.29, 'eval_runtime': 301.9326, 'eval_samples_per_
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{'loss': 0.4511, 'grad_norm': 5.01708984375, 'learning_rate': 8.325581395348837e-06, 'epoch': 2.22}

{'loss': 0.3808, 'grad_norm': 6.198825359344482, 'learning_rate': 7.395348837209303e-06, 'epoch': 2.54}

{'loss': 0.3425, 'grad_norm': 5.0866875648498535, 'learning_rate': 6.465116279069767e-06, 'epoch': 2.86}
```
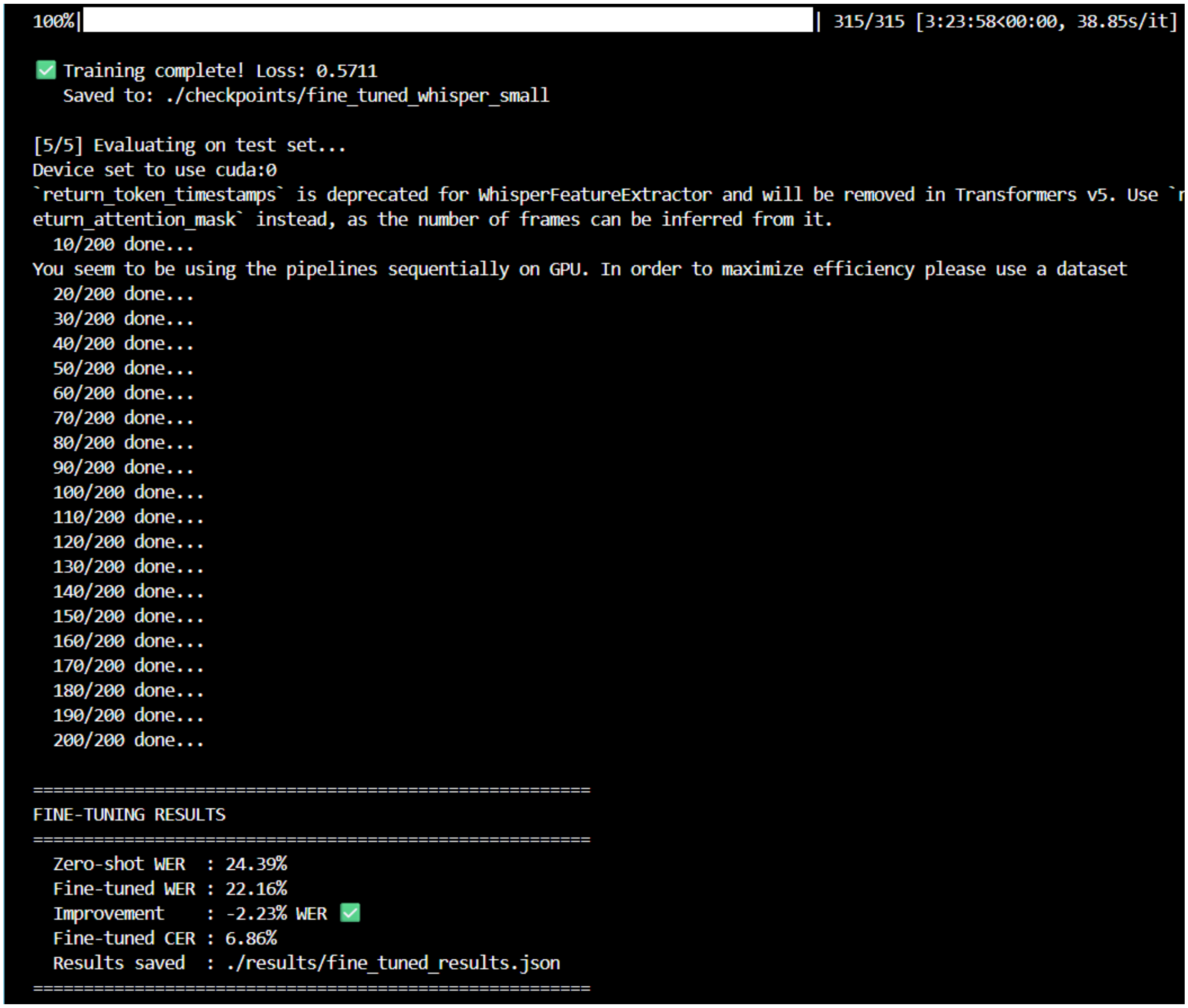

```
{'eval_loss': 0.43383410573005676, 'eval_wer': 23.72, 'eval_cer': 7.89, 'eval_runtime': 303.618, 'eval_samples_per
_second': 0.659, 'eval_steps_per_second': 0.165, 'epoch': 3.0}

 62%|█████████████████████████████████████████████                  | 195/315 [2:05:18<1:40:00, 50.01s/
it]
 62%|███████████████████████████████████████████                | 196/315 [2:05:56<1:31:56, 46.35s/it]
{'loss': 0.283, 'grad_norm': 4.574166774749756, 'learning_rate': 5.534883720930233e-06, 'epoch': 3.18}
 66%|█████████████████████████████████████████████              | 207/315 [2:12:25<1:08:36, 38.12s/it]
```



```
100%|████████████████████████████████████████████████████████████████████████| 315/315 [3:23:58<00:00, 38.85s/it]

✅ Training complete! Loss: 0.5711
   Saved to: ./checkpoints/fine_tuned_whisper_small

[5/5] Evaluating on test set...
Device set to use cuda:0
`return_token_timestamps` is deprecated for WhisperFeatureExtractor and will be removed in Transformers v5. Use `r
eturn_attention_mask` instead, as the number of frames can be inferred from it.
  10/200 done...
You seem to be using the pipelines sequentially on GPU. In order to maximize efficiency please use a dataset
  20/200 done...
  30/200 done...
  40/200 done...
  50/200 done...
  60/200 done...
  70/200 done...
  80/200 done...
  90/200 done...
  100/200 done...
  110/200 done...
  120/200 done...
  130/200 done...
  140/200 done...
  150/200 done...
  160/200 done...
  170/200 done...
  180/200 done...
  190/200 done...
  200/200 done...

=======================================================
FINE-TUNING RESULTS
=======================================================
  Zero-shot WER  : 24.39%
  Fine-tuned WER : 22.16%
  Improvement    : -2.23% WER ✅
  Fine-tuned CER : 6.86%
  Results saved  : ./results/fine_tuned_results.json
=======================================================
```